

Improving User Experience in VR-Based Gamma Entrainment During Cognitive Tasks via Edge-Only Flickering: A Pilot Study

Ayoung Shin^{*†}, Haechan Lee^{*‡}, and Uijong Ju[§]

Department of Information Display, Kyung Hee University

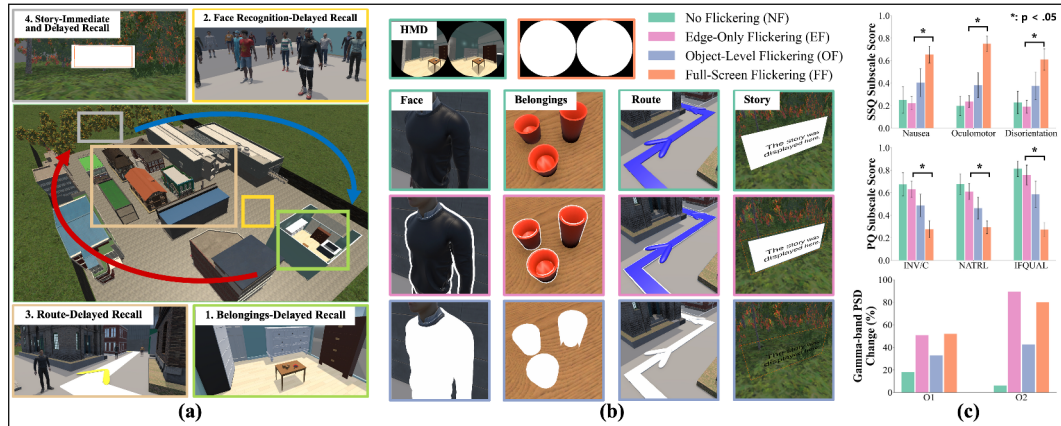


Figure 1: (a) Bird's-eye view of the VR-based RBMT-III environment, where each colored box represents a scene corresponding to a subtest. (b) Example images illustrating task-relevant objects under different conditions, with the border color corresponding to the legend. In FF, the entire display flickers, whereas in EF and OF, the appearance alternates at 40 Hz between the NF state and the condition-specific colored box. (c) From top to bottom, subscale scores of SSQ and PQ across flickering conditions, followed by the gamma-band PSD changes. Each bar color corresponds to the legend, and the error bars indicate the standard error.

ABSTRACT

Gamma-frequency entrainment is a potential therapeutic approach for Alzheimer's disease. This study developed a method integrating gamma entrainment into a VR-based cognitive task while preserving user experience. A pilot study with eight participants evaluated user experience after completing four Rivermead Behavioral Memory Test-III (RBMT-III) subtests under different flickering conditions. Results showed that the Edge-Only flickering condition, closely resembling the no-flicker condition, yielded better user experience. EEG analyses confirmed higher gamma entrainment in this condition. These findings suggest that Edge-Only flickering effectively induces gamma entrainment while maintaining user experience.

Index Terms: Virtual Reality, Gamma ENtrainment Using Sensory stimulation, User Experience, Cognitive Task, RBMT-III

1 INTRODUCTION

Gamma ENtrainment Using Sensory stimulation (GENUS) has recently been investigated as a promising therapeutic approach for Alzheimer's disease by inducing gamma entrainment through sensory stimulation. Prior studies suggest that the effectiveness of this approach might depend on the degree of gamma entrainment and

can be enhanced when 40 Hz stimulation is combined with cognitive tasks [1].

One recent study showed that it is feasible to implement GENUS within a virtual reality (VR) environment [2]. This suggests the possibility that immersive cognitive tasks can be performed simultaneously with 40 Hz gamma entrainment in VR. Compared with other approaches, VR offers greater accessibility, enabling personalized and continuous daily training and assessment.

However, VR-based gamma entrainment using visual flickering may induce visual discomfort and diminish user experience. To address this issue, the present pilot study investigates how different flickering areas within the visual display affect user experience. The study aims to identify an optimal flickering strategy that induces strong gamma entrainment while maintaining a positive user experience, thereby enabling sustained engagement.

2 EXPERIMENT

2.1 Participants

A total of 8 participants (6 male; mean age = 25.25 years, SD = 2.66, range = 21–29) were recruited. All participants were university students, highly experienced in VR, with no history of neurological disease.

2.2 Apparatus

The experiment was conducted on a PC equipped with a Nvidia GeForce RTX 4090. The VR experimental environment was created by Unity3D 2022.3.8f1. The Varjo XR-4 headset (3840 × 3744 resolution per eye, 120° FOV) was utilized with the dedicated controllers. An Emotiv EPOC X (14-channel, 256 Hz sampling rate) electroencephalography (EEG) headset was employed to record neural responses for assessing gamma entrainment.

^{*}These authors equally contributed to this work.

[†]e-mail: sinay2002@khu.ac.kr

[‡]e-mail: chanchans0905@khu.ac.kr

[§]e-mail: juuijong@khu.ac.kr (corresponding author)

2.3 Experimental design

The cognitive task was based on the Rivermead Behavioral Memory Test-III (RBMT-III) and implemented in a VR environment in which instructions were provided to allow all participants to perform navigation according to the task flow. The environment consisted of three areas: **Room** (10 × 10 m), **Street** (100 × 60 m, 13 intersections), and **Garden** (tree-enclosed area) [3].

Four RBMT-III subtests were selected for this study (see Fig. 1-(a)): **Belongings-Delayed Recall**, which assessed memory for object-drawer associations involving two household items; **Route-Delayed Recall**, which required memorization of a route sequence while following a blue guideline from the Room to the Garden; **Face Recognition-Delayed Recall**, which assessed memory for the appearances of 15 virtual individuals; and **Story-Immediate and Delayed Recall**, which involved reading a short story in the Garden followed by both immediate and delayed recall.

To identify an optimal flickering method, we implemented four different flickering conditions, which were defined as follows (see Fig. 1-(b)): **No Flickering (NF)** — No flickering was provided. **Full-Screen Flickering (FF)** — The entire VR display flickered. **Object-Level Flickering (OF)** — The entire surface of the task-relevant object flickered. **Edge-Only Flickering (EF)** — Only the edges of the task-relevant object flickered. The flickering was implemented using custom shaders and was briefly verified at 40Hz using a CdS-based light-dependent resistor and an Arduino Uno.

2.4 Procedure

Participants completed one experimental session per day over four consecutive days. During the first visit, they received a detailed explanation of the experimental procedures. Each participant experienced all flickering conditions once in a counterbalanced order. To minimize learning effects, task-relevant objects—including belongings, individuals, routes, and stories—were varied across sessions.

At the beginning of each session, participants departed from the Room after memorizing object-drawer associations and navigated toward the Garden while memorizing task-relevant information for each subtest (following the red arrow in Fig. 1-(a), from subtest 1 to 4). Participants then returned from the Garden to the Room, recalling the memorized information and completing the corresponding cognitive tasks (following the blue arrow in Fig. 1-(a), from subtest 4 to 1). Each session lasted approximately 20 minutes.

2.5 Measurements

As the primary focus of this study was on user experience and gamma entrainment rather than task performance, measurements targeted subjective questionnaires and EEG-based metrics. To evaluate user experience under different flickering conditions, the Simulator Sickness Questionnaire (SSQ) was administered to assess Nausea, Oculomotor, and Disorientation, and the Presence Questionnaire (PQ) was used to measure Involvement/Control (INV/C), Natural (NAT), and Interface Quality (InQ) [4, 5].

2.6 EEG Validation

EEG validation was conducted separately to verify gamma entrainment while minimizing non-neural artifacts via EEG preprocessing. A gray screen (baseline) and a flickering-applied RBMT-III interface were viewed for 5 minutes each. Gamma-band (30–45 Hz) EEG responses were analyzed as baseline-normalized changes in integrated power spectral density (PSD). The O1 and O2 electrodes were selected for the strongest occipital entrainment at 40 Hz [1].

3 RESULTS

Due to the small sample size ($N = 8$) and the non-normal distribution of the data, we applied the Friedman test to investigate differences in subjective experiences across flickering conditions. The results showed that the flickering condition significantly affected

all SSQ subscales (Nausea: $\chi^2(3) = 11.466$, $p = .009$; Oculomotor: $\chi^2(3) = 16.676$, $p < .001$; Disorientation: $\chi^2(3) = 11.838$, $p = .008$). Next, post hoc pairwise comparisons using the Wilcoxon signed-rank test revealed that EF had significantly lower scores than FF across all subscales (Nausea: $Z = -2.527$, $p = .012$; Oculomotor: $Z = -2.524$, $p = .012$; Disorientation: $Z = -2.533$, $p = .011$).

Next, we applied the Friedman test to investigate differences in presence across flickering conditions. The results showed that the flickering condition significantly affected all PQ subscales (INV/C: $\chi^2(3) = 16.050$, $p = .001$; NAT: $\chi^2(3) = 11.026$, $p = .012$; InQ: $\chi^2(3) = 15.943$, $p = .001$). Post hoc analyses using the Wilcoxon signed-rank test revealed that EF had significantly higher scores than FF across all subscales (INV/C: $Z = -2.527$, $p = .012$; NAT: $Z = -2.313$, $p = .021$; InQ: $Z = -2.524$, $p = .012$).

3.1 EEG results

EEG recordings from the occipital electrodes (O1 and O2) showed increased baseline-normalized gamma-band PSD under the FF, OF, and EF conditions, indicating gamma entrainment. Notably, the EF condition produced gamma entrainment comparable to FF and greater than OF. Quantitatively, gamma-entrainment changes at O1 were 17.9% (NF), 50.8% (EF), 32.8% (OF), and 51.9% (FF), whereas at O2 they were 6.12% (NF), 89.4% (EF), 42.4% (OF), and 79.9% (FF). The analysis results are summarized in Fig. 1-(c).

4 CONCLUSIONS AND FUTURE WORK

This study showed that user experience in VR-based cognitive tasks is influenced by the visual flickering and suggests that the EF condition may provide an optimal balance between gamma entrainment and user experience. Reducing flickering area from full-screen to edges increased presence and reduced sickness and visual discomfort, indicating that user experience is sensitive to flickering extent.

Notably, Edge flickering on the task-relevant object did not disrupt task-related visual processing, resulting in the most favorable post-hoc questionnaire outcomes for the EF condition. Gamma entrainment was minimal in the NF condition, while all flickering conditions (FF, OF, and EF) elicited greater gamma activity than NF. Among them, EF showed a higher gamma-band PSD change from baseline, suggesting relatively effective gamma entrainment.

Overall, although this study is limited by its small sample size, the findings suggest a promising direction for integrating gamma entrainment with VR-based cognitive tasks. Future research should aim to expand the participant pool, examine the long-term effects on cognitive task performance, and validate this approach in individuals with mild cognitive impairment, ultimately contributing to advances in VR-based gamma entrainment for Alzheimer's disease.

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